

A Simplified Tool for Testing of Feature Selection and Classification Algorithms in Motor Imagery of Right and Left Hands of EEG Signals

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Abstract—Some algorithms or a combination of them are more appropriated than others depending on the type of data that is being analyzed and what features and parameters are being used. In the analysis of motor imagery (MI) in an offline EEG-based brain-computer interface (BCI), different codes with different parameters are often used, making it harder to compare the effects of the algorithms applied. In this paper, we propose a simplified and limited tool that aims to aid in the testing of feature extraction, selection and classification algorithms separately or combined for the analysis of motor imagery in offline EEG-based brain signals while providing some information about the intermediate steps of a BCI construction. A known data set is used in order to ease the comparison between other researches. Only data from channels C3, Cz and C4 are used and the MI of left hand and right hand are analyzed. The data is filtered using a band-pass Chebyshev type II filter between 5 and 35Hz. Then, The rhythms mu and beta are isolated using a discrete wavelet transform (DWT) algorithm with a db4 mother wavelet of level 5. The proposed system has two outputs: the coefficients of the DWT related to the rhythms mu and beta; and a feature vector with three chosen features that can be used as an input to a classifier. The features extracted are mean, variance and energy. These are simple but effective features. Fixing some of the parameters simplifies the tool, offers a better environment for comparison of algorithms and allows the user to focus on specific steps of a BCI construction such as the feature selection and the classification phases.

Index Terms—BCI, EEG, DWT, feature extraction, feature selection, classification, motor imagery, SMR, ERD/ERS, simple tool

I. INTRODUCTION

The advances in hardware and software have enabled many studies in the field of brain-computer interface (BCI) in the last decades [1]–[8]. In a BCI system, a user with the aid of an interface can control an external device using only brain signals, without the help from any peripheral nerves or muscles [1].

Due to its simplicity and low cost compared to other methods of brain function analysis, the most used signal acquisition technique in BCI studies is the electroencephalogram (EEG)

[2]. The EEG is a noninvasive method that uses electrodes on the surface of the scalp placed at specific locations to monitor and record electrical activity signals generated by the firing of a group of neurons in the brain. EEG-based BCIs have been used for communication and control to help people suffering from severe motor disabilities. Both the realization of a movement and the imagination of the realization of the movement, also known as motor imagery (MI), produce signals that behave in a similar manner [1]–[3]. This is relevant because it enables a person who is unable to execute the movement to simply imagine its realization. An EEG can monitor MI over the sensorimotor cortex to observe the behavior of sensorimotor rhythms (SMR), such as the mu and beta rhythms. One of the main reasons why SMR has been used in BCIs is that users are capable of learning how to control MI features voluntarily after a certain amount of training [6].

According to [1] the biggest challenge in a BCI system for robotic and prosthetic devices is how to provide control signals that are fast, accurate and reliable. One of the reasons for that is because the processing of brain signals is hampered by its properties. Brain signals are known for being nonstationary, nonlinear and prone to noise [9]. Furthermore, EEG signals are different from one user to another in their frequency variations and task observations [2]. Moreover, these signals can be altered by properties like tiredness and emotional changes. Hence the necessity to build models with algorithms that can adapt to each user. Finding better features and better algorithms for feature extraction, selection and classification, can help the achievement of more accurate results, making the system more reliable. However, in [10] the authors observed a lack of comparisons between classifiers within the same study using identical parameters. Ideally, classifiers should be tested in the same context, that is, with the same users, using the same feature extraction methods, protocol and parameters [1]. Similarly, this concept can be spread to all of the algorithms present in the system so that more suitable conclusions about the best algorithms for this type of signal can be found.

There are some tools available to help with the construction

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of a BCI [11]–[14]. Yet, BCI research and development is an interdisciplinary problem, involving professionals from neuroscientists to computer engineers [6]. And even though these tools usually offer functions and algorithms ready to be used, it is still necessary to find out what are the outputs of these functions and how they can be used as input to the next steps of the development. Sometimes, it is required to understand not only the theory behind the algorithm but also its implementation. As a result, these tools present a certain degree of difficulty in their learning and use by people whose expertise are not exactly related to some of the terminologies and concepts used. Besides, they allow a certain freedom in the choice of data sets, algorithms and type of analysis. This freedom is only good to some extent, since it ends up bringing a lot of variability, making it harder to compare the algorithms efficiently.

This paper intends to provide a simplified tool for the testing of feature extraction, selection and classification algorithms separately or combined for the analysis of motor imagery of left and right hands using SMR in offline EEG-based brain signals within a limited environment. The purpose of this limitation is to enable the comparison of the implemented algorithms with maximum similarity and minimum variability, keeping the data set, method of frequency isolation and preprocessing fixed under the same context. All of these while providing some detail about the intermediate steps of the construction of the system by analyzing its partial results. The goal of this work is to enable the discovery of systems that can achieve the best results in velocity, precision and reliability of control signals. The proposed tool uses a discrete wavelet transform (DWT) algorithm to separate the desired sensorimotor signals and build a simple feature vector that can be used as input to classification algorithms.

II. ERD/ERS OF SENSORIMOTOR RHYTHMS

Sensorimotor rhythms are a common type of control signal used in EEG-based BCI of motor imagery. SMR are electrophysiological signals that can be monitored noninvasively with electrodes located over the sensorimotor cortex. According to the ISO 10-20 distribution pattern, this area is under the channels $C3$, Cz and $C4$ [15].

SMR comprises the rhythms mu (μ) and beta (β), which are oscillations in the brain activity located in the frequency bands $8Hz$ to $13Hz$ and $13Hz$ to $30Hz$, respectively [6]. However, a small variation on these frequency ranges might be encountered in other studies.

The rhythms mu and beta can be used as an indication of movement because the amplitude of these rhythms presents changes when movement, preparation of movement or motor imagery occur. With movement, the amplitude of the rhythm mu decreases significantly. After movement and with relaxation, its amplitude increased again. A decrease in the amplitude is called event-related desynchronization (ERD), while an increase in the amplitude is known as event-related synchronization (ERS). Neither ERD nor ERS require actual movement, they also occur with MI [6].

For hand movement, μ ERD happens 1-2 seconds before movement onset, reaching the minimum value during movement. While β ERS reach its maximum value 1-2 seconds after the end of the movement [8]. ERD/ERS are contralateral to the movement. When MI of the right hand happens, the left side of the motor and sensory cortex of the brain displays ERD, while the right side displays ERS and the opposite happens when MI of the left hand occurs [7]. As a result, usually the channel $C4$ is monitored when observing movements from the left hand side and channel $C3$, for movements from the right hand side [16].

It is possible to isolate the frequency bands corresponding to the mu and beta rhythms with the use of simple filters or with the help of algorithms, such as the discrete wavelet transform.

III. DISCRETE WAVELET TRANSFORM

Nonstationary signals, such as biological signals, are signals whose frequencies vary in time. For the analysis of EEG signals in terms of mu and beta rhythms, it is necessary to extract the frequency bands present in the signal as well as when in time they happen. For this reason, the Fast Fourier Transform (FFT) is not an appropriated method for analyzing such signals, since it gives information only in the frequency domain [2], [17]. The Wavelet Transform (WT) on the other hand, gives information in both time and frequency domain. A variant of the WT called Discrete Wavelet Transform (DWT) provides sufficient information both for analysis and synthesis of the original signal. Moreover, it is considerably easier to implement and provides less redundant information as far as the reconstruction of the signal goes compared to another type of WT, the Continuous Wavelet Transform (CWT) [17].

The DWT decomposes the signal into approximation and detail information and analyses it at different frequency bands with different resolutions using scales and translations of a window function known as the *mother wavelet*. This window is shifted through the signal in varied scales. In other words, the translations can be understood as the shifting of the window in a time period and the scale as the width of the window at each moment. The signal in time domain is passed through a number of high-pass and low-pass filters depending on the desired level of decomposition. The resultant signals have both higher and lower frequencies. Higher frequencies are best determined in time domain and lower frequencies are enhanced by frequency [2]. After the filtering, according to the Nyquist's rule, half of the samples can be eliminated since the highest frequency becomes half of the value that it was before. Therefore, the signal is downsampled by two [17]. The level of decomposition depends on the sampling frequency and the desired frequency bands.

The approximation and the detail coefficients with their respective bandwidths, can be roughly calculated as depicted in Table I, where f_s is the sample frequency.

It is important to choose the correctly wavelet family and the number of decomposition levels of the DWT when analyzing EEG signals. The *Daubechies* wavelet family is known for its orthogonality property and efficient filter implementation

TABLE I
EXAMPLE OF COEFFICIENTS AND THEIR BANDWIDTHS

Approximation coefficients	BW	Detail coefficients	BW
A1	0 - fs/4	D1	fs/4 - fs/2
A2	0 - fs/8	D2	fs/8 - fs/4
A3	0 - fs/16	D3	fs/16 - fs/8
A4	0 - fs/32	D4	fs/32 - fs/16
A5	0 - fs/64	D5	fs/64 - fs/32

[18]. Moreover, it has been pointed out that the Daubechies of order 4 (db4) is suitable for identifying changes in EEG signals [2], [18]. An orthogonal transform means that the energy in the signal is preserved and partitioned among the coefficients [19]. The energy of the frequency bands are different with the imagination of distinctive tasks. After movement, the energy in $C3$ and $C4$ are different [18]. Thus, with the use of an orthogonal wavelet, the energy can be used as a feature to compose the feature vector.

IV. EXPERIMENT

The proposed tool receives a file of recorded EEG signals as input and gives as output either just the coefficients that represent the mu and beta rhythms extracted with the use of a DWT or a feature vector formed by the features mean, variance and energy of these coefficients. This way, the user can choose arbitrary features to be extracted or use an algorithm to do so. And if the goal is to test just classification algorithms, then the user can simply use the given feature vector. The data set, preprocessing phase, rhythm isolation and feature extraction are always carried in the same manner so that the algorithms of feature selection and classification can be tested and compared. All of the algorithms are implemented using MATLAB® and the biosig toolbox [13].

The data set used is from the Graz BCI Competition IV, 2008 - 2a. It is a public, well known and extensively used data set that consists of EEG data for motor imagery of four different tasks from nine subjects. A trial is a recording of one of the tasks and has the duration of approximately 9 seconds. A run consists of 48 trials, being 12 trials for each of the tasks. A file is recorded for each subject and it contains one session, which comprises 6 runs, yielding a total of 288 trials per session. 22 electrodes were used for the recording of the brain activity and their placement were carried based on the ISO 10-20 system. The signals were sampled with 250 Hz and were filtered with a band-pass filter between 0.5Hz and 100Hz and with a 50Hz notch filter enabled to suppress line noise. For the recording, at $t = 2s$, that is, after 2 seconds, a cue in the form of an arrow pointing up, down, left or right (corresponding to each of the imagined movements) was displayed for 1.25s, indicating that the subject should begin the imagination and keep it going until $t = 6s$. More details about this data set can be found in the description file [20].

In this paper, only the activity of two tasks were analyzed, namely the motor imagery of left and right hands. As a result,

each run is composed of 24 trials and the total of trial per session is 144. In order to simplify and reduce redundant information, only channels $C3$, Cz and $C4$ were used. Indeed, since the intention is to observe ERD/ERS from mu and beta rhythms, which are rhythms associated with motor imagery, it is reasonable to use only the electrodes placed over the sensorimotor cortex. After acquiring the signals from the chosen channels, the trials were separated according to the instruction from the description of the competition [20]. The epoch extraction phase takes into consideration the events of ERD/ERS. Considering the average reaction time of a health person as 500ms [16], the movement occurs around 4s and for this reason the epochs chosen are from 2-7s so that both events can be observed.

Filtering is done using a band-pass Chebyshev type II filter from 5-35Hz, which comprises the range of mu (8-12Hz) and beta (18-30Hz) rhythms and help avoid artifacts. The DWT algorithm is used to isolate the mu and beta rhythms from the filtered data. The mother wavelet used is the db4 of level 5. As seen in Table I, since the sample frequency is 250Hz the estimated frequency bands for the detail coefficients of each level, from level 1 to level 5, are 62.50 - 125.00Hz; 31.25 - 62.50Hz; 15.62 - 31.25Hz; 7.81 - 15.62Hz; and 3.90 - 7.81Hz. The detail coefficients $D3$ and $D4$ are the ones with the correspondent ranges of beta and mu rhythms, respectively. Hence, these two coefficients are one of the outputs of the tool and are the ones used for the extraction of information for the montage of the feature vector. Here, the extracted features are *mean*, *variance* and *energy*. These features were calculated according to (1), (2) and (3), respectively. Where N is the total number of samples and x_i is the current sample.

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (1)$$

$$\sigma^2 = \frac{1}{N-1} \sum_{i=1}^N |x_i - \bar{x}|^2 \quad (2)$$

$$E = \sum_{i=-\infty}^{\infty} |x_i|^2 \quad (3)$$

Even though it is possible to extract other kinds of features, these features are commonly used with this type of signals and other works show that they have efficient results [18]. One of the advantages of using the DWT is the capability of having information in both time and frequency domains. For this reason, the features were chosen so they could represent information in both domains. For example, the mean represents information in the time domain while the energy represents information in the frequency domain.

V. RESULTS

According to [21], variations on mu and beta rhythms are best seen contralateral to the movement. Thus, movement of the right hand, is best observed in channel $C3$, which is located in the left hemisphere, and movement of the left hand, is

best observed in channel $C4$, which is located in the right hemisphere.

Fig. 1 shows a scalogram plotted using the coefficients of the CWT that represents the frequency bands of mu (8-12Hz) and beta (18-30Hz) observed over the $C3$ channel for MI of the right hand. The CWT was used to enhance the visual aspect of the signal's behavior. But for all of the calculations mentioned in the previous section, the DWT is used. As mentioned before, movement onset happens around $t = 4s$ and the epoch extracted is from 2-7s. Also, μ ERD reaches its minimum value during movement and β ERS reach its maximum value 1-2 seconds after the end of the movement. In Fig. 1, the darker the color the lower the amplitude. This way, μ ERD can be observed from $t = 3s$ to $t = 4s$ and β ERS from $t = 4.5s$ to $t = 5.5s$.

VI. CONCLUSION

The main intention of the proposed tool is to ease the comparison and testing of selection and classification algorithms of motor imagery activities of hands in EEG-based BCIs using SMR, within a limited environment. The signal processing is done up to the step of frequency isolation, leaving the implementation of the subsequent phases of feature extraction, selection and classification to the users. Because it is a simplified tool, users that are initiating their studies in the field of BCI may find it easier to understand it and use it. In addition, users that want to focus mainly on the testing of classification algorithms don't have to deal with the previous steps of signal processing. This reduction on the complexity of BCI systems' design could help with the adhesion of specialists and their deeper and more specific research in their own fields, providing greater speed in the development of more suitable systems. Similarly, finding better features and algorithms decrease the error rate of the system, increasing its precision and reliability. Regarding the intermediate steps'

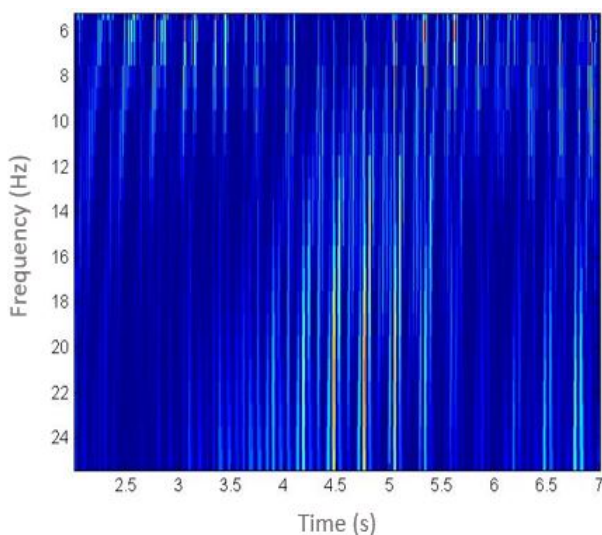


Fig. 1. A scalogram of mu and beta rhythms of all of the samples of a user over the $C3$ channel related to motor imagery of the right hand.

results, the scalogram shows that they are in accordance with the theory described and, therefore, can also be used to help in the understanding of the system. Tests using this tool can be done to find the algorithms that lead to the best possible accuracy in the prediction of MI activities for the designed system. A high percentage of accuracy is important to increase the usability of the system in a more practical way.

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